Numerical processing of financial data
APM 50442 EP
Part 3: from sequential data to portfolio selection

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2025-2026

1 Sequential stock data in one dimension

2 Sequential stock data in higher dimension

3 From sequential data to portfolio selection
- Evaluating the ex-post performance of a portfolio
- Building an optimal portfolio: Markowitz approach
- Introduction to statistical arbitrage

4 Market microstructure

1 Sequential stock data in one dimension

2 Sequential stock data in higher dimension

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  - Risk-adjusted performance criteria
  - Risk and performance attribution
- Building an optimal portfolio: Markowitz approach
- Introduction to statistical arbitrage

4 Market microstructure

From performance to risk-adjusted performance

A game proposed by Bernoulli (1713)

A fair coin is tossed. If heads appears, the coin is tossed again until tails appears. The player wins 2 euros if tails appears on the first toss, 4 euros on the second, ... 2^n euros on the n-th toss. How much would you pay to take part to this game?

- We know that the price is the expected value of the discounted future cash flows. $\rightsquigarrow$ What is the price of Bernoulli's game?
- **St. Petersburg paradox**: no one would pay such a price.
- One way to go beyond the paradox? Let's consider **risk aversion** and utility function, which biases our perception of the gain.

Risk measures used in portfolio management

- Volatility: $\sigma(X) = \sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}$
- Lower semi-deviation, focused on downside risk: $\sigma_-(X) = \sqrt{\mathbb{E}[(X - \mathbb{E}[X])_-^2]}$, where $[X]_- = \max\{0, -X\}$.
- Value at risk, more useful in banking regulation: $\mathrm{Var}_\alpha(X) = F_X^{-1}(1-\alpha)$, where F_X is the cdf of X.

Risk-adjusted performance ratio – The Sharpe ratio

Définition

Let R be the (total) return of a portfolio for a given investment horizon, R^b the return of a benchmark, and R^f the risk-free rate.
The **Sharpe ratio** *(William Sharpe, 1966) is the ratio of the expected risk premium over its standard deviation:*

$$Sharpe = \frac{\mathbb{E}(R - R^f)}{\sqrt{\mathbb{V}ar(R - R^f)}}.$$

The **Information ratio** *is the Sharpe ratio of the returns in excess of a benchmark (excess return over tracking error volatility):*

$$IR = \frac{\mathbb{E}(R - R^b)}{\sqrt{\mathbb{V}ar(R - R^b)}}.$$

- In ex-post evaluation, expectation and variance are evaluated empirically on historical data.
- These ratios are sensitive to the time scale: the annualized ratio is obtained by multiplying by $\sqrt{260}$ if the returns are daily returns, with 260 days in a year.

Risk-adjusted performance ratio – The Sharpe ratio

The higher the Sharpe, the better. The portfolio manager can achieve this by increasing the return or decreasing the volatility (considered here as the risk measure).

Some drawbacks

- The ratio may be negative, but in this case it is difficult to interpret it and the portfolio manager could ameliorate it by increasing the risk (perverse effect).
- Returns and volatility are implicitly considered as having the same magnitude but the comparison of Sharpe ratios between different strategies is not as obvious:
  - it should depend on the type of asset (what meaning for money-market with almost zero volatility?)
  - it should depend on the rebalancing frequency and thus on the skill of the portfolio manager (fundamental law of active management, by Grinold and Kahn).
- A positive jump implies a rise of volatility and thus an increased risk and a lower Sharpe, whereas this jump is something positive. Possibility to separate *positive and negative volatility*?
- Estimation difficulties: very unstable, depends on time scale, etc.

Asymetric risk-adjusted performance ratio

- **Sortino ratio**, for a minimal acceptable return $r^\star$ (say equal to zero): $S(r^\star) = \frac{\mathbb{E}[r - r^\star]}{\sqrt{\mathbb{E}[\min(r - r^\star, 0)^2]}}$,
- **Upside potential ratio** (Sortino, van der Meer, Plantinga, 1999), upside performance relative to downside risk: $U(r^\star) = \frac{\mathbb{E}[\max(r - r^\star, 0)]}{\sqrt{\mathbb{E}[\min(r - r^\star, 0)^2]}}$.
- **Omega ratio** (Keating and Shadwick, 2002), for F the cdf of returns:

$$\Omega(r^\star) = \frac{\int_{r^\star}^{+\infty} (1 - F(r))\, dr}{\int_{-\infty}^{r^\star} F(r) dr}.$$

$\rightsquigarrow$ Show that, if $r^\star = 0$, we get:

$$\Omega(0) = \frac{\mathbb{E}[\max(R, 0)]}{\mathbb{E}[-\min(R, 0)]}.$$

A generalized risk-adjusted performance ratio

The risk measure focuses on:

- the partial first moment, for Ω,
- the second moment, for the Sharpe ratio,
- the partial second moment, for S and U.

Could be generalized to other (partial) moments. In particular, asymmetry is better described by skewness.

Définition

The n-th lower partial moment function of a random variable of cumulative distribution function F is $\mathcal{M}_n(\tau) = \int_{-\infty}^{\tau} (\tau - r)^n dF(r)$.

The **Kappa ratio** (van Harlow, 1991) is:

$$K_n(r^\star) = \frac{\mathbb{E}[r - r^\star]}{\sqrt[n]{\mathcal{M}_n(r^\star)}}.$$

As particular cases:

- $S(r^\star) = K_2(r^\star)$,
- $\Omega(r^\star) = 1 + K_1(r^\star)$ (by integrations by parts).

Replacing partial moments by maximum drawdown

Définition

For a series of prices $P_1, ..., P_n$, the **maximum drawdown** *is the maximal loss (a positive metric) encountered at one of the dates of the observed time interval, regarding another date in the same interval:*

$$MDD = \max_{1 \leq i < j \leq n} -\left(\frac{P_j - P_i}{P_i}\right).$$

Also known as peak to valley loss.

How to calculate it rapidly?

- For $j = 2$ to n, calculate the drawdown at the current date j from the previous peak: $m_j = -\left(\frac{P_j}{\max\limits_{1 \leq i < j} P_i} - 1\right)$.
- The maximum drawdown is the maximum of these drawdowns: $MDD = \max\limits_{2 \leq j \leq n} m_j$.

Related metrics:

- the **maximum drawdown duration**: $j - i$ such that $MDD = -\left(\frac{P_j - P_i}{P_i}\right)$,
- the **recovery duration** (if it exists): if $[\![i, j]\!]$ is the MDD time interval, the recovery duration is $k - j$ such that k is the lowest integer greater than j such that $P_k > P_i$ (then $\frac{P_k - P_j}{P_i} > MDD$, even if $P_k = P_i$).

Replacing partial moments by maximum drawdown

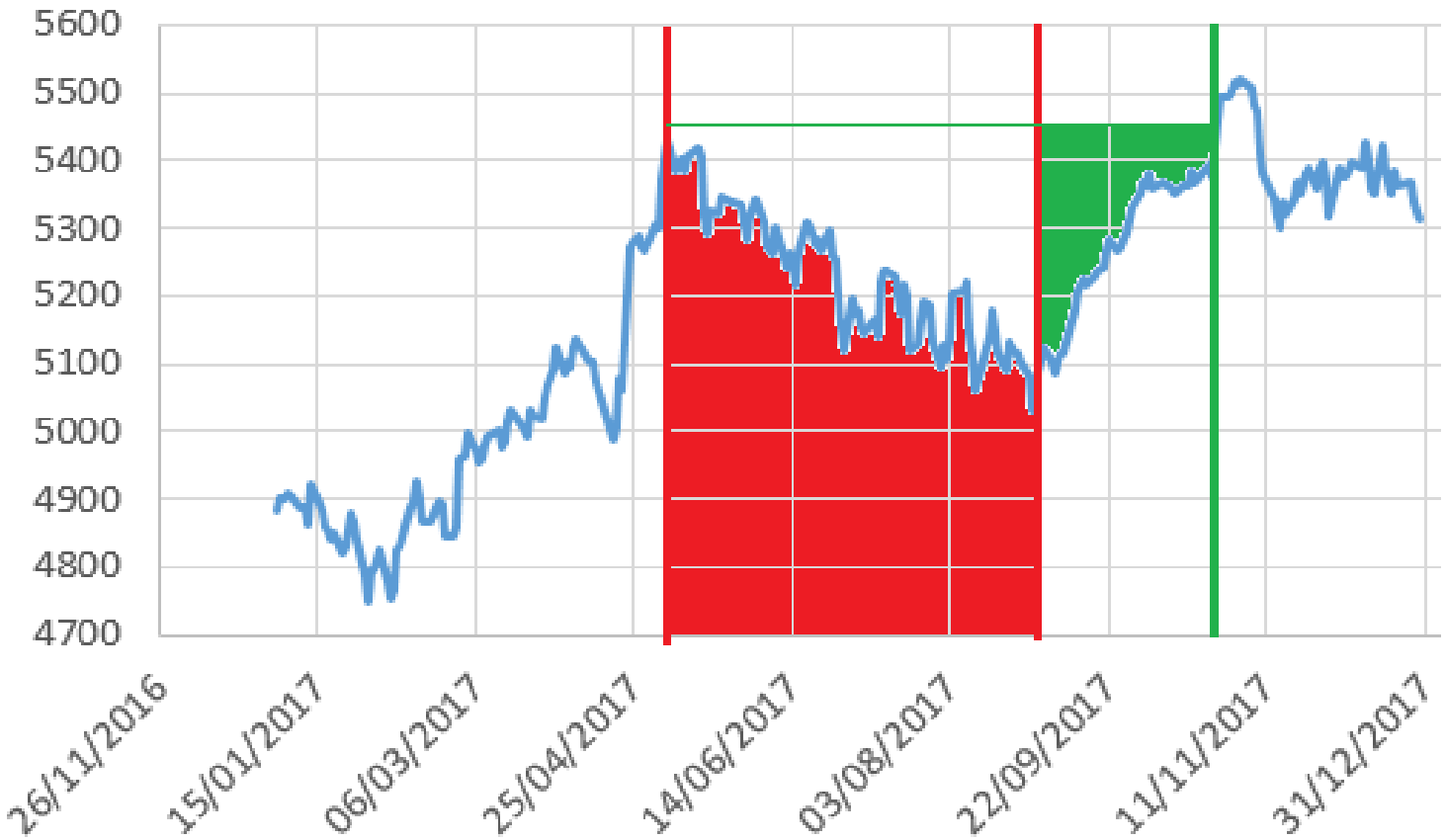

Figure: Maximum drawdown in 2017 for the CAC40 index: 7.37% between 8th May and 29th August (81 trading days), for a recovery above the price of 5442.10 the 27th October (43 trading days).

Replacing partial moments by maximum drawdown

The **Sterling ratio** is an adaptation of Sharpe or Sortino ratios.

Warning concerning the time scale!

- In Sharpe (or Sortino), it is easy to calculate metrics at the numerator and the denominator on the same time scale, for example average daily return and daily volatility or lower partial volatility.
- Calculating daily MDD is meaningless (unless working in high frequency, and in this case the daily MDD is a proxy of the lower partial realized volatility).
- We prefer decomposing the observation time interval in adjacent subsets of same duration (for example calendar years) and calculate the average of the MDD of each subset.
- To be consistent, the numerator must be an average of the return on the same subsets.

For example, the yearly Sterling ratio, calculated on N years, is:

$$\text{Sterling} = \frac{\frac{1}{N}\sum_{i=1}^{N} R_i}{\frac{1}{N}\sum_{i=1}^{N} \text{MDD}_i},$$

where $R_i = \frac{P_{256i}}{P_{256(i-1)}} - 1$ and $\text{MDD}_i = \max_{256(i-1)\leq m<n\leq 256i} - \left(\frac{P_n - P_m}{P_n}\right)$.

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4 Market microstructure

Contribution of an asset to performance and risk

In a time interval $[s, t]$ (all returns between s and t, weights at time s), the price return of a portfolio is decomposed in:

$$R = \sum_i w_i r_i,$$

with weight w_i and return r_i for asset i.

(Additive) contribution of asset i

- to the performance: $w_i r_i$,
- to the volatility:

$$\frac{w_i \sigma(r_i) \sum_j w_j \rho(r_i, r_j) \sigma(r_j)}{\sigma(R)},$$

with $\rho(X, Y)$ the correlation between X and Y.
$\rightsquigarrow$ Why such a formula?

$\rightsquigarrow$ Python exercise, Question 2.

Performance attribution

For a benchmarked portfolio, two objectives: good stock picking (like in previous page) and good sector allocation (new!). Decomposition of the excess return of the portfolio as

$$R - \sum_{j=1}^{N} w_j^{b,\text{sector}} r_j^{b,\text{sector}} = \sum_{j=1}^{N} \Big\{ \underbrace{(w_j^{\text{sector}} - w_j^{b,\text{sector}}) r_j^{b,\text{sector}}}_{\text{sector allocation}} + \sum_{i \in \mathcal{S}_j} \underbrace{w_i (r_i - r_j^{b,\text{sector}})}_{\text{stock picking}} \Big\},$$

where

- $\mathcal{S}_j$ is the set of assets in sector j,
- N the number of sectors,
- $w_j^{b,\text{sector}} = \sum_{i \in \mathcal{S}_j} w_i^b$ the weight of the sector j in the benchmark,
- and $r_j^{b,\text{sector}} = \sum_{i \in \mathcal{S}_j} \frac{w_i^b}{w_j^{b,\text{sector}}} r_i$ the return of the sector j in the benchmark.

Performance attribution

Equivalently, noting $\phi(i)$ the sector of stock i (in a universe of n stocks), the performance of the portfolio is the sum of:

- the performance of the trader in allocating sectors: $\sum_{i=1}^{n} (w_i - w_i^b) r_{\phi(i)}^{b,\text{sector}}$.
- the performance of the trader in selecting stocks: $\sum_{i=1}^{n} w_i (r_i - r_{\phi(i)}^{b,\text{sector}})$.

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2 Sequential stock data in higher dimension

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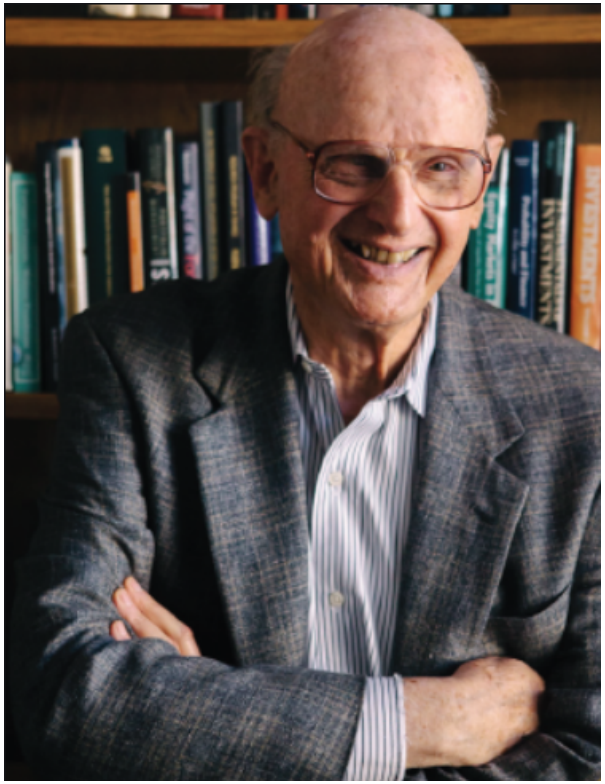
4 Market microstructure

Modern portfolio theory (MPT)

Idea of MPT introduced by Markowitz (1927-2023) in 1952: use engineering methods to build a portfolio; instead of a fundamental approach, choose securities and weights as a solution to a constrained optimization problem.



A long time before recognition:

> *"When I defended my dissertation as a student in the Economics Department of the University of Chicago, Professor Milton Friedman argued that portfolio theory was not Economics, and that they could not award me a Ph.D. degree in Economics for a dissertation which was not in Economics. I assume that he was only half serious, since they did award me the degree without long debate. As to the merits of his arguments, at this point I am quite willing to concede: at the time I defended my dissertation, portfolio theory was not part of Economics. But now it is."*

(Harry Makowitz, Nobel lecture, December 7, 1990)

Modern portfolio theory

Maximize a utility consisting in a risk-adjusted performance

In t, for a time horizon h, find weights $\Omega_t^\star = (\omega_{1,t}^\star, ..., \omega_{N,t}^\star)$ maximizing a utility in which the risk is considered to be the variance of the portfolio (returns are in excess of the risk-free rate):

$$\Omega_t^\star = \underset{(\omega_1,\ldots,\omega_N)\in\mathbb{R}^N}{\operatorname{argmax}} \sum_{j=1}^{N} \omega_j \mathbb{E}[r_{j,t,t+h}] - \lambda \mathbb{V}\mathrm{ar}\left[\sum_{j=1}^{N} \omega_j r_{j,t,t+h}\right].$$

Analytic solution, noting $\mathbf{r} = (r_{1,t,t+h}, ..., r_{N,t,t+h})'$, $\mu = \mathbb{E}[\mathbf{r}]$, $\Sigma = \mathbb{V}\mathrm{ar}[\mathbf{r}]$:

$$\Omega_t^\star = \frac{1}{2\lambda}\Sigma^{-1}\mu.$$

The parameter λ is the risk aversion. How to choose it?

- With the $\sum_{i=1}^{N} \omega_i = 1$ constraint (we may also require $\forall i \in [\![1, N]\!], \omega_i \in [0, 1]$), but not really risk aversion... $\rightsquigarrow$ What is $\Omega^\star$ then?
- We can also see λ as the Lagrange multiplier of the problem of maximizing the expected return under the constraint of reaching a given level of portfolio variance.
- By modifying λ, we describe a set of portfolios that constitute the **efficient frontier**.

Modern portfolio theory – A Lagrangian viewpoint

Maximizing the performance

- Maximize the performance for a given level of risk (instead of risk aversion λ):

$$\max_{\Omega} \Omega'\mu \qquad \text{such that} \qquad \Omega'\Sigma\Omega = \alpha^2.$$

- Equivalent to maximizing a Lagrangian function:

$$\max_{\Omega,\gamma} \mathcal{L}_1(\Omega,\gamma) \qquad \text{where} \qquad \mathcal{L}_1(\Omega,\gamma) = \Omega'\mu - \gamma\left(\Omega'\Sigma\Omega - \alpha^2\right).$$

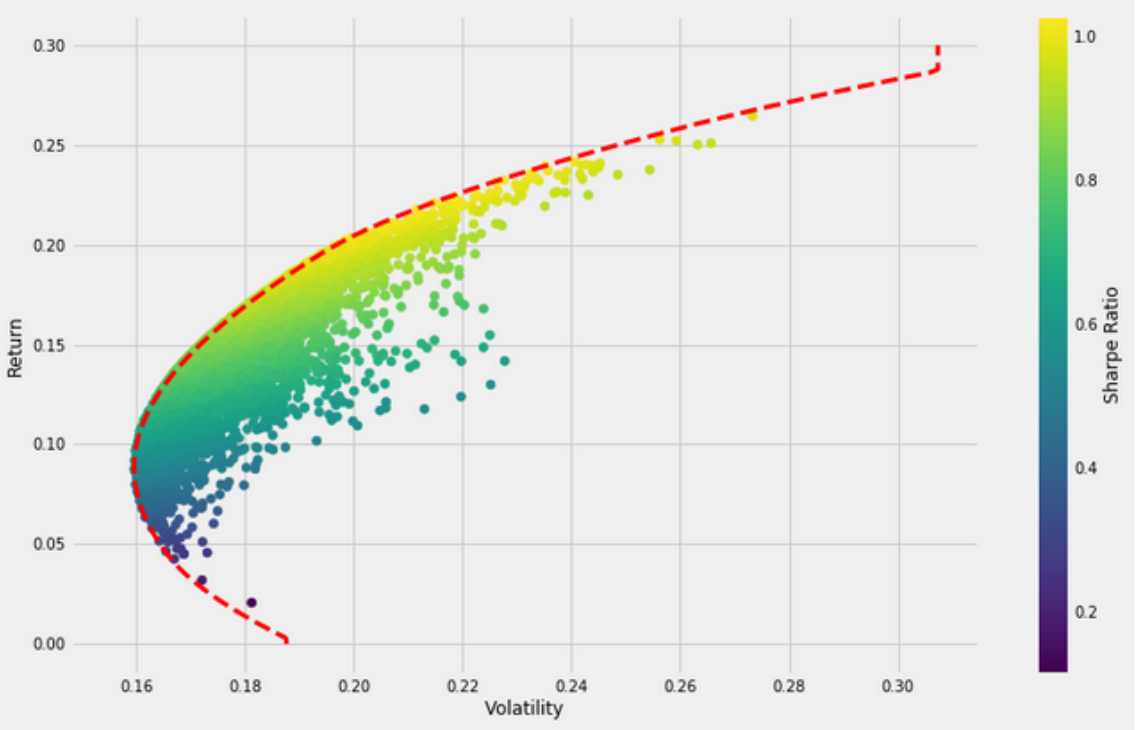
- $\rightsquigarrow$ Show that the solution of the Lagrangian problem is the following:

$$\Omega^\star = \frac{1}{2\gamma^\star}\Sigma^{-1}\mu \qquad \text{and} \qquad \gamma^\star = \frac{1}{2|\alpha|}\sqrt{\mu'\Sigma^{-1}\mu}.$$

Minimizing the risk

- $\min_{\Omega} \Omega'\Sigma\Omega \qquad \text{such that} \qquad \Omega'\mu = \xi.$
- $\min_{\Omega,\delta} \mathcal{L}_2(\Omega,\delta) \qquad \text{where} \qquad \mathcal{L}_2(\Omega,\delta) = \Omega'\Sigma\Omega - \delta\left(\Omega'\mu - \xi\right).$
- $\Omega^\star = \frac{\delta^\star}{2}\Sigma^{-1}\mu \qquad \text{and} \qquad \delta^\star = \frac{2\xi}{\mu'\Sigma^{-1}\mu}.$

Modern portfolio theory – The efficient frontier



We recognize:

- the minimal-variance portfolio,
- many suboptimal portfolios (inefficient).

⇝ How does our optimal portfolio evolve with λ?
⇝ How does diversification work?
⇝ Python exercise, Question 1.

Modern portfolio theory – Extensions

What expected return and what variance to consider?

- These quantities are not observed, even *a posteriori*.
- Simplest solution: replace with historical estimates. But will a security that has had good performances in the past have good performances in the future?
- A model of the evolution of returns is needed: CAPM and factor models or econometric approach (models for jumps, asymmetry, heteroscedasticity, ...).
- Expectation and variance are two statistics of a distribution: importance of correctly estimating the (conditional) probability densities of returns.

Extensions of Markowitz theory:

- Addition of no leverage ($\sum_i |\omega_i| \leq 1$) or sector constraints (e.g. pharmaceutical industry stocks must represent between 5% and 10% of the portfolio): inequality constraints $\longrightarrow$ Karush-Kuhn-Tucker problem.
- Uncertain horizon h.
- Other criteria to optimize, notably for risk.
- Taking into account transaction costs.
- Taking into account the need to have a parsimonious portfolio.

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4 Market microstructure

Biases in backtesting

1. **Look-ahead bias**: in-sample performance, overfitting...
2. **Survivorship bias**: appears when working with a static universe, whereas reality is dynamic, like the constituents of an index.
3. **Liquidity bias**: for each transaction, there are costs, slippage, permanent/transitory impact. $\Rightarrow$ Making a transaction will modify the time series of prices and transaction may not be executed at the displayed price.
4. **Lack of statistical significance**:
   - A single trajectory is anecdotal.
   - Split the dataset in independent test sets (e.g. performance per month), representing various market regimes.
   - Draw randomly the investment universe in a large one (and repeat): is the result sensitive to this selection?
   - Much more difficult: simulate (an unlimited number of)*realistic* trajectories.
   - Statistical tests (possibly a binomial test) to validate the strategy.
   - The average performance is less informative than a (conditional) distribution of performances.

How to evaluate the pdf of simulated performances?

An estimator $\hat{f}$ of a density should check: $\forall x \in \mathbb{R}, \hat{f}(x) \geq 0$, and $\int_{-\infty}^{+\infty} \hat{f}(x)dx = 1$. From a series of observed price returns $X_1, ..., X_n$, the probability density could be estimated by:

- A **histogram** (not continuous, not invariant by translation), with a discretization $a_1 < ... < a_m$: if $x \in [a_i, a_{i+1})$, then

$$\hat{f}(x) = \frac{1}{n(a_{i+1} - a_i)} \sum_{j=1}^{n} \mathbb{1}_{[a_i, a_{i+1})}(X_j).$$

- A **kernel density**, where the Kernel K, a density with cdf $\mathcal{K}$ can be, for example, $K(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$, with a smoothing parameter $h > 0$ indicating the smoothness/robustness:

$$\boxed{\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right)} \text{ and } \hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{K}\left(\frac{x - X_i}{h}\right).$$

  Drawback: fixed-size resolution (h), thus spurious effects for tails and smoothing of essential elements in the main part of the distribution.

- An **adaptive (or variable) kernel density**:

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{hd_{i,k}} K\left(\frac{x - X_i}{hd_{i,k}}\right) \text{ and } \hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{K}\left(\frac{x - X_i}{hd_{i,k}}\right),$$

  where $d_{i,k}$ is the Euclidean distance between X_i and its k-th nearest neighbour among the $n - 1$ other observations, $X_1, ..., X_{i-1}, X_{i+1}, ..., X_n$.

How to evaluate the pdf of simulated performances?

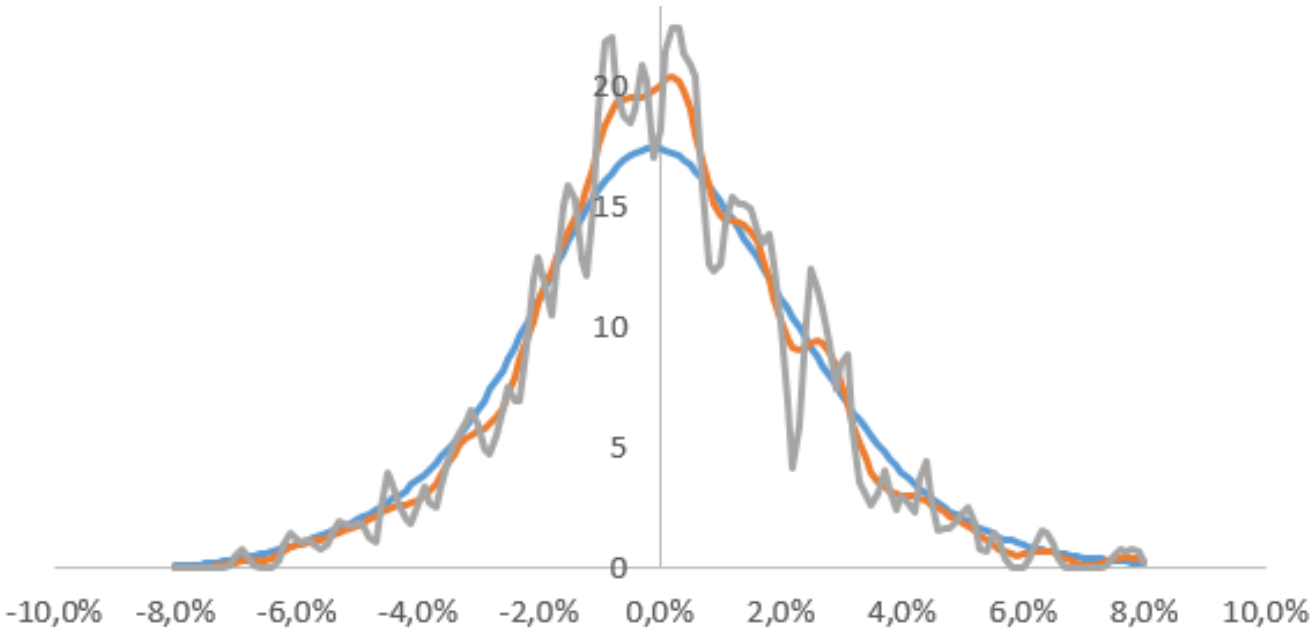

Figure: Kernel density of historical daily price returns (Natixis stock, 2015-2016) for $h = 0.01$ (smooth one), $h = 0.003$, and $h = 0.001$ (roughest one).

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4 Market microstructure

Standard arbitrage strategies

1. **Cross-market arbitrage**: An asset traded on several exchanges will not have the same price everywhere; arbitrage when best bid quote in one market is higher than its lowest offer quote in another market (mainly HFT).
2. **Momentum**: Take profit from positive serial dependence; for example long position on the stock with the best performance over the last 3 months and short position on that with the worst performance.
3. **Pairs trading**: Invest in two strongly correlated assets:
   - Find β such that the price spread $P_A - \beta P_B$ is stationary.
   - Stationary $\Rightarrow$ mean-reverting, so if the spread strongly positively deviates from zero, sell A and buy βB (market-neutral position), until the spread is back to a value close to zero.
4. **Post-earnings-announcement drift**: new information is not immediately digested by market; drift during several weeks in the direction of the earning surprise $\Rightarrow$ basically be long when positive announcement.

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The Hurst exponent

Definition

A (discretized) fractional Brownian motion (fBm) B_H of Hurst exponent $0 < H < 1$ and scale parameter σ^2, observed at times $0 = t_0 < \ldots < t_n$, is such that:

- $B_H(t_0) = 0$,
- $\mathbb{E}\{B_H(t_i)\} = 0$, for all i,
- $B_H(t_j) - B_H(t_i) \sim \mathcal{N}(0, \sigma^2 |t_j - t_i|^{2H})$.

Interpretation:

- if $H = 1/2$, the fBm is identical to a standard Brownian motion,
- if $H > 1/2$, the increments are persistent, presence of long memory,
- if $H < 1/2$, the increments are anti-persistent.

Application to statistical arbitrage

1. If a price spread is an fBm with $H < 1/2$: pairs trading.
2. If a log-price is an fBm: use serial dependence to forecast future variations.

$\rightsquigarrow$ How to estimate H?

Forecasting with the Hurst exponent

Focusing on a single lag (but can be generalized to a higher number of lags). $R_{t,t+h} = X_{t+h} - X_t$ predicted from $R_{t-\delta,t} = X_{t-\delta} - X_t$. The predictor is noted $\hat{R}_{t,t+h|\delta}$ and, consistently with the covariance approach, is of the form $\beta R_{t-\delta,t}$, where β is defined by:

$$\beta = \frac{\mathrm{Cov}(R_{t,t+h}, R_{t-\delta,t})}{\mathbb{V}\mathrm{ar}(R_{t-\delta,t})} = \frac{1}{2}\left[\left(\frac{h}{\delta}+1\right)^{2H} - \left(\frac{h}{\delta}\right)^{2H} - 1\right].$$

Théorème

When predicting with $\beta R_{t-\delta,t}$ the increment $R_{t,t+h}$ of an fBm of Hurst exponent $H \in (0,1/2) \cup (1/2,1)$, where $h, \delta > 0$ are the increment durations, the hit ratio is:

$$\rho = \mathbb{P}\left[\hat{R}_{t,t+h|\Delta} R_{t,t+h} \geq 0\right] = 1 - \frac{1}{\pi}\arctan\left(\sqrt{\frac{1}{\beta^2}\left(\frac{h}{\delta}\right)^{2H} - 1}\right).$$

Forecasting with the Hurst exponent

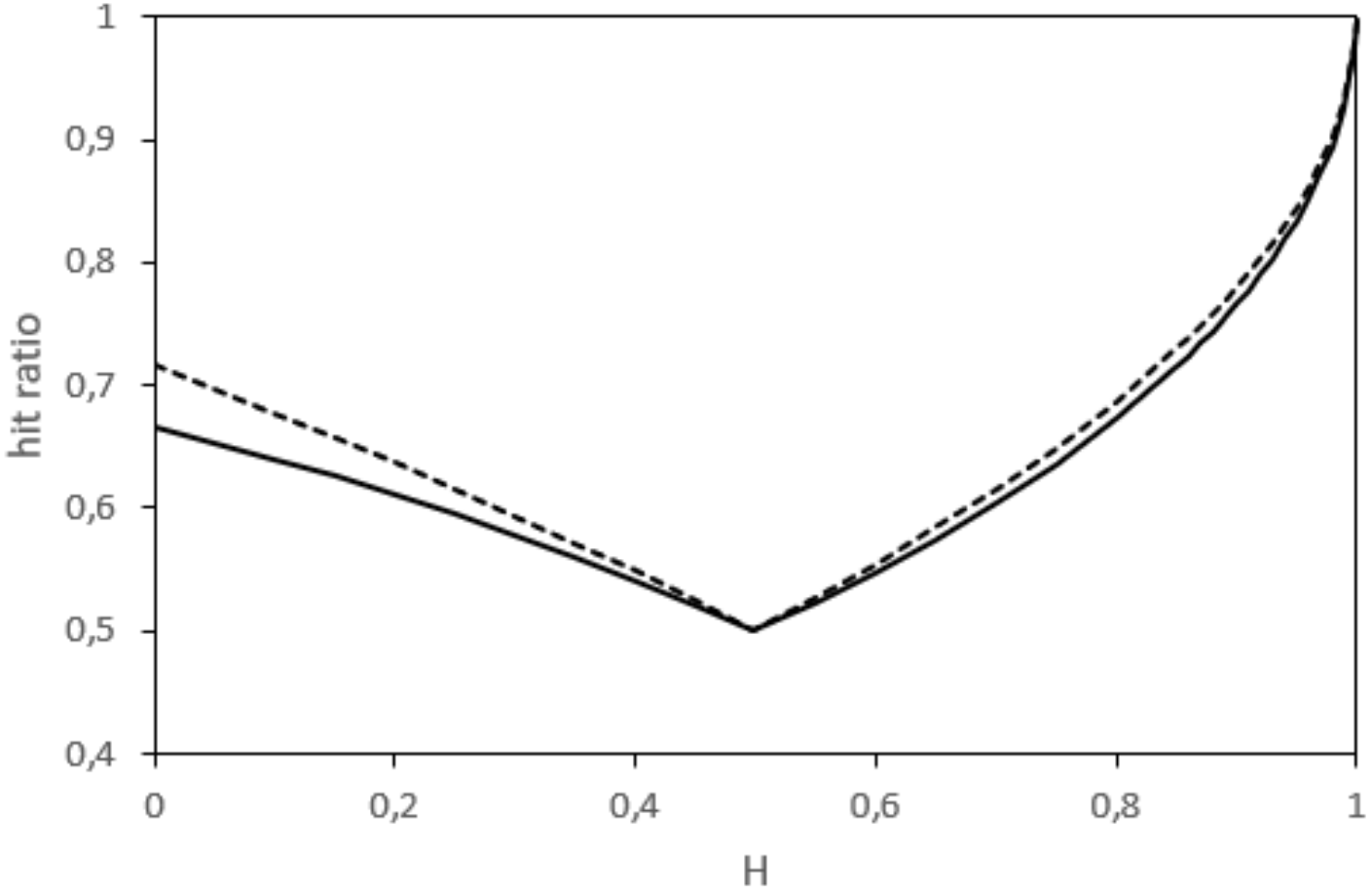

Figure: Hit ratio with $\delta = h$ with one time lag (solid line) and 5 time lags (dotted line).

To invest or not to invest?

Intuition

Investment decisions based on predictions close to zero are more risky than those based on larger predictions. The hit ratio is less relevant than a risk-adjusted performance.

Using the thresholding function $f_\theta : x \in \mathbb{R} \mapsto x\mathbb{1}_{\{|x| \geq \theta\}}$, the predictor is now $f_\theta\left(\hat{R}_{t,t+h|\Delta}\right)$.

Définition

Non-conditional probability of good sign forecast: $p^+(\theta) = \mathbb{P}\left[f_\theta\left(\hat{R}_{t,t+h|\Delta}\right) R_{t,t+h} > 0\right]$,
of bad sign forecast: $p^-(\theta) = \mathbb{P}\left[f_\theta\left(\hat{R}_{t,t+h|\Delta}\right) R_{t,t+h} < 0\right]$,
of zero forecast: $p^0(\theta) = \mathbb{P}\left[f_\theta\left(\hat{R}_{t,t+h|\Delta}\right) = 0\right]$.

In the fBm case, $p^0(\theta) = -1 + 2N\left(\frac{\theta}{\beta\delta^H\sigma}\right)$.

Risk-adjusted performance

The trading strategy: buying a fixed quantity of the asset if one predicts a positive price return, short selling the same quantity if one predicts a negative return, not investing if the expected return of the asset is below the threshold θ in absolute value.
Return of the strategy:

$$R_{t,t+h}^{\text{strat}}(\theta) = \begin{cases} 0 & \text{if } f_\theta\left(\hat{R}_{t,t+h|\Delta}|\right) = 0 \\ R_{t,t+h} & \text{if } f_\theta\left(\hat{R}_{t,t+h|\Delta}|\right) > 0 \\ -R_{t,t+h} & \text{if } f_\theta\left(\hat{R}_{t,t+h|\Delta}|\right) < 0. \end{cases}$$

- Expected return: $\widetilde{R}_{t,t+h}(\theta) = \mathbb{E}[R_{t,t+h}^{\text{strat}}(\theta)]$.
- Average lower absolute semi-deviation risk measure: $\widetilde{\sigma}_{t,t+h}^{-}(\theta) = -\mathbb{E}\left[\min(0, R_{t,t+h}^{\text{strat}}(\theta))\right]$.
- Risk-adjusted performance: $\widetilde{R}_\lambda(\theta) = \widetilde{R}_{t,t+h}(\theta) - \lambda\widetilde{\sigma}_{t,t+h}^{-}(\theta)$.

Risk-adjusted performance

Théorème

Let X_t be an fBm of parameters $H \in (0, 1/2) \cup (1/2, 1)$ and $\sigma > 0$. Let $h > 0$ and $\hat{R}_{t,t+h|\Delta}$ be a predictor of $R_{t,t+h}$ based on a single past return. Given $\theta \geq 0$, we have:

$$\widetilde{R}_{t,t+h}(\theta) = 2ag\left(\frac{\theta}{a}\right)$$

and

$$\widetilde{\sigma}^{-}_{t,t+h}(\theta) = -2aN\left(-\frac{\theta}{b}\right)g\left(\frac{\theta}{a}\right) + \sqrt{\frac{2}{\pi}}\sigma h^H N\left(-\theta\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}\right),$$

where $a = \beta\delta^H\sigma$, $b = \sigma\sqrt{h^{2H} - \beta^2\delta^{2H}}$, and g is the Gaussian probability density function.

Risk-adjusted performance

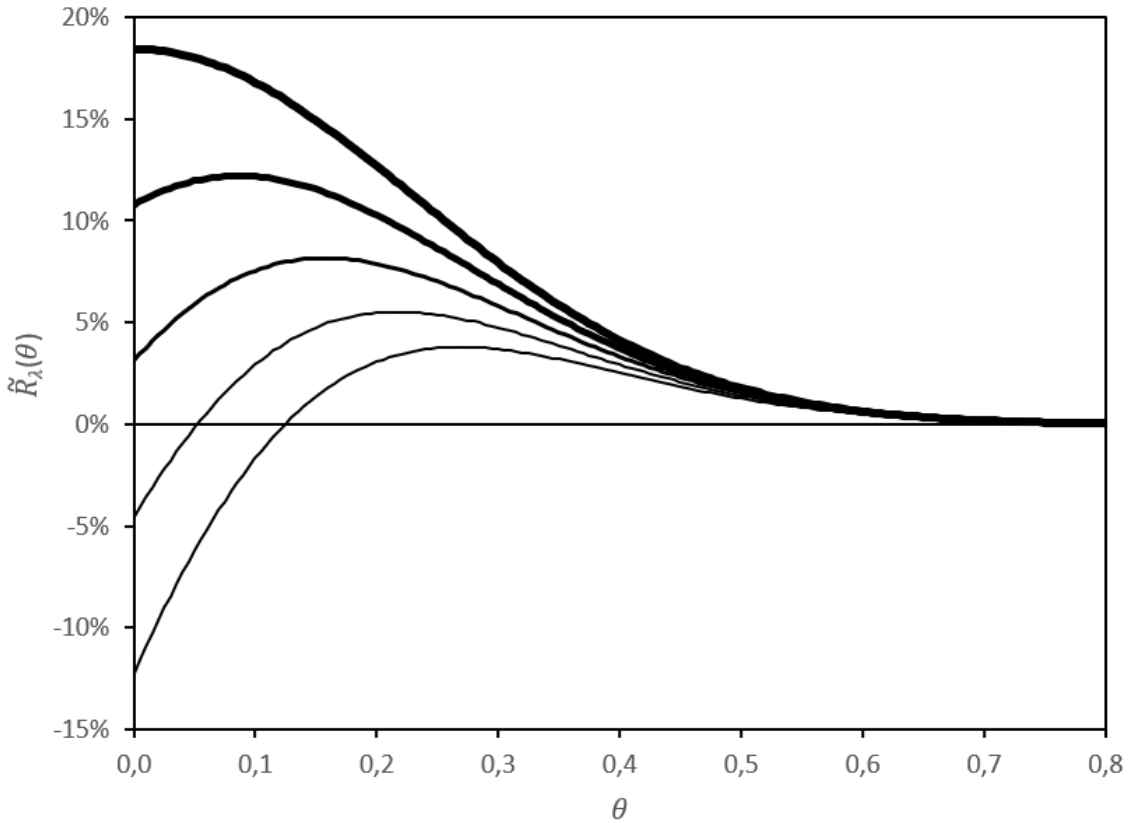

Figure: Theoretical $\widetilde{R}_\lambda(\theta)$ as a function of the threshold θ, with risk aversion $\lambda = 0$, 0.25, 0.5, 0.75, 1 (fatter to thinner). Forecast horizon $h = 1$, $n = 1$ lagged returns in the predictor, fBm with $H = 0.65$ and $\sigma = 1$.

Optimal threshold

$$\theta_\lambda^\star = \underset{\theta \geq 0}{\operatorname{argmax}}\ \widetilde{R}_\lambda(\theta)$$

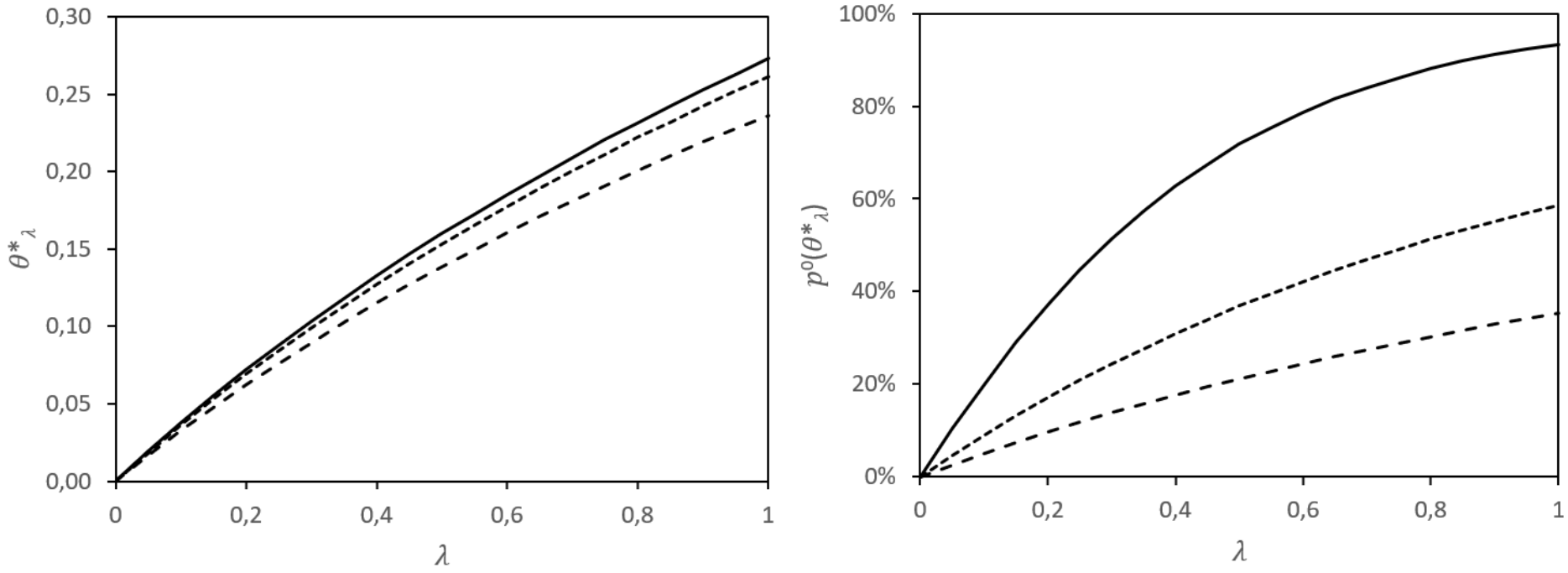

Figure: Theoretical optimal threshold $\theta_\lambda^\star$ (left) and corresponding non-conditional zero forecast probability $p^0(\theta_\lambda^\star)$ (right) for an fBm with $\sigma = 1$ and $H = 0.6$, 0.7, and 0.8 (from top to bottom). Forecast horizon $h = 1$, $n = 1$ lagged returns in the predictor.

Optimal threshold

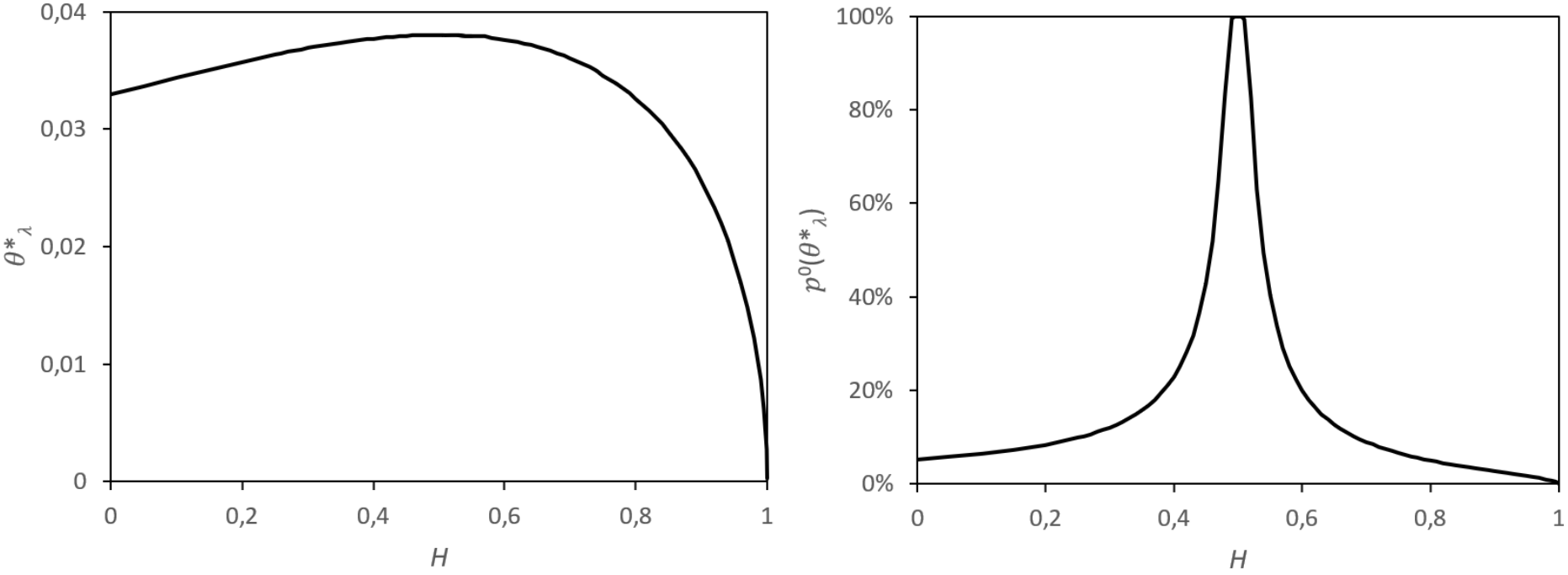

Figure: Theoretical $\theta^\star_\lambda$ (left) and corresponding $p^0(\theta^\star_\lambda)$ (right) when risk aversion is $\lambda = 0.1$.

- Time series of FX prices with a one-minute sampling: EURGBP, EURUSD, GBPUSD.
- Focus on one week in order to limit the effects of closing markets: from the 23rd June 2019 to the 28th June 2019.
- Estimation of Hurst exponent in 12-hour windows, updated every minute.
- Forecast of the evolution of the log-price at a one-hour horizon, with two lagged returns:
  - **Naive lags**: one-hour and two-hour past returns.
  - **Ex ante optimal lags**: For example, when $H = 0.35$, we consider the last 21-minute and 176-minute returns.

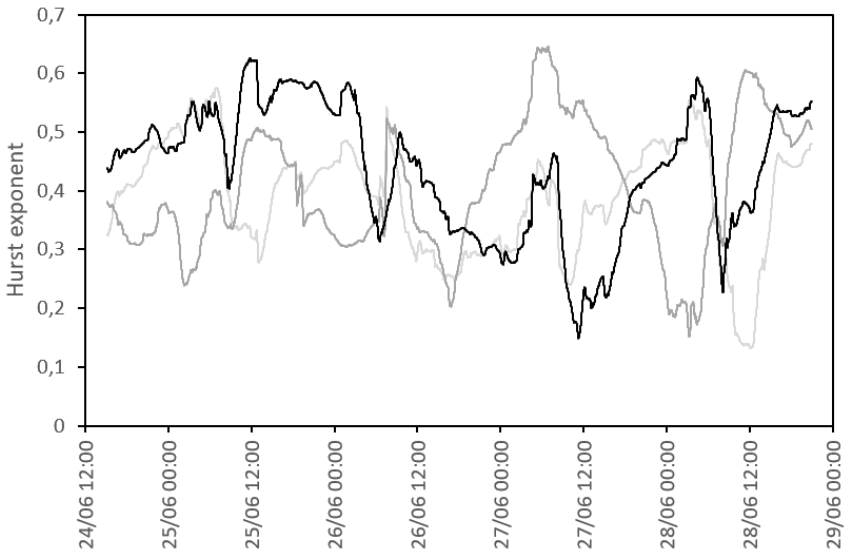

Figure: Estimated time-varying Hurst exponent of the log-prices of EURGBP (black), EURUSD (dark grey), GBPUSD (light grey).

FX rate	Method	Return	Risk	$\widetilde{R}_\lambda(\theta)$	$\widehat{p^+}$	$\widehat{p^-}$	$\widehat{p^0}$
EURGBP	Naive lags, $\theta = 0$	-0.51	2.47	-2.97	49.8%	50.2%	0.0%
	Optimal lags, $\theta = 0$	3.66	2.26	1.40	52.9%	47.1%	0.0%
	Optimal lags and θ	3.09	1.50	1.59	30.2%	26.7%	43.1%
EURUSD	Naive lags, $\theta = 0$	2.82	1.85	0.97	54.6%	45.4%	0.0%
	Optimal lags, $\theta = 0$	1.52	1.91	-0.39	54.3%	45.7%	0.0%
	Optimal lags and θ	1.37	1.20	0.17	34.4%	29.6%	36.0%
GBPUSD	Naive lags, $\theta = 0$	1.97	2.42	-0.45	50.6%	49.4%	0.0%
	Optimal lags, $\theta = 0$	4.22	2.31	1.91	52.4%	47.6%	0.0%
	Optimal lags and θ	4.58	1.63	2.96	33.5%	28.8%	37.7%

Table: Empirical average one-hour return, risk (lower absolute semi-deviation), and risk-adjusted return (with $\lambda = 0.1$) multiplied respectively by 10^5, 10^4, and 10^5. As soon as $\theta = 0$, $\widehat{p^+}$ is simply the hit ratio.

End of Part 3.